

GYAN SETU

Chapter-1, Number System Notes

1. What is a Polynomial?

A polynomial is an expression made up of variables (like x , y) and constants (like numbers), combined using addition, subtraction, and multiplication.

Examples of Polynomials:

- $2x + 3$ (Linear)
- $x^2 - 3x + 2$ (Quadratic)
- $x^3 - x^2 + 2x + 1$ (Cubic)

NOTE: Not Polynomials: $1/x$, $\sqrt{x}$, because powers must be whole numbers.

2. Types of Polynomials by Degree

- **Degree 1:** Linear (e.g., $2x + 3$)
 - **Degree 2:** Quadratic (e.g., $x^2 - 3x + 2$)
 - **Degree 3:** Cubic (e.g., $x^3 + x - 1$)
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3. What are Zeroes of a Polynomial?

A zero of a polynomial is the value of x which makes the polynomial equal to 0.

Example: For $p(x) = x^2 - 3x - 4$, $p(4) = 0$ and $p(-1) = 0$.

So, zeroes are 4 and -1.

4. Graphical Meaning of Zeroes

- Zeroes are the points where the graph cuts the x-axis.
 - A quadratic polynomial can have 2, 1, or 0 zeroes based on how the graph touches x-axis.
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5. Relationship between Zeroes and Coefficients

Quadratic Polynomial:

For $ax^2 + bx + c$, if α and β are zeroes, then:

- Sum of zeroes $= \alpha + \beta = -b/a$
- Product of zeroes $= \alpha\beta = c/a$

Example: $x^2 + 7x + 10$

- Factor $= (x + 2)(x + 5)$, so zeroes $= -2$ and -5
- Sum $= -2 + (-5) = -7 = -b/a$
- Product $= (-2)(-5) = 10 = c/a$

Cubic Polynomial

For $ax^3 + bx^2 + cx + d$, zeroes α, β, γ :

- $\alpha + \beta + \gamma = -b/a$
 - $\alpha\beta + \beta\gamma + \gamma\alpha = c/a$
 - $\alpha\beta\gamma = -d/a$
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6. Summary

- Linear: 1 zero
- Quadratic: At most 2 zeroes
- Cubic: At most 3 zeroes
- Zeroes = x-values where graph touches x-axis
- Use formulas to find sum/product of zeroes easily.

